Early and adequate antibiotic therapy in hospital-onset sepsis: a target trial emulation applying machine learning on real-world data

Scott A. Cohen[1,2], Inyoung Jun[2], Sarah Ser[2], J. Glenn Morris Jr.[1], Mattia Prosperi[2]

[1]Emerging Pathogens Institute; [2] Department of Epidemiology
College of Medicine & Public Health and Health Professions, University of Florida

Data Intelligence Systems Lab
Department of Epidemiology
UNIVERSITY of FLORIDA

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INTRODUCTION

BACKGROUND

- Antimicrobial resistance (AMR) is a leading public health concern, particularly in the context of hospital-acquired infections.
- Delays in appropriate antibiotics increase morbidity and mortality, however studies assessing burden and prognostic impact of antibiotic therapy in this critically ill population are limited.
- Given the limitations of randomized trials, innovative strategies using real-world data and computational methods are needed.

OBJECTIVE:

- To leverage a combination of target trial emulation and machine learning to estimate causal effects of early and adequate antibiotic therapy in hospital-onset sepsis, as well as the moderating role or AMR

METHODS

- Retrospective cohort study designed with the **target trial emulation framework** using data from UF Integrated Data Repository (IRB201900652).
- Approximately 140,000 hospital encounters with documented bacterial infection from 2010 to 2020 were examined from eligibility for adult patients who were hospitalized hospital-onset sepsis. (**Figure 1)**

Target Trial Specification

Eligibility Criteria →	Adult patients hospitalized for at least 48 hours with clinical suspicion of new onset of sepsis based on US CDC Definitions
Treatment Groups →	Receipt of antibiotic regimen within 6, 12, 18, 24, 36 or 48 hours of blood culture collection that: 1. **Covers the pathogen via imputed AST*** 2. **Does not cover pathogen via imputed AST***
Assignment/ Randomization →	Randomization emulated using baseline and time-varying covariates within 6, 12, 18, 24, 36 or 48 of culture
Follow-Up Period →	Blood culture collection → 30 days or discharge or death
Outcome →	In-hospital mortality within 30 days
Causal Contrast →	Observational analog Per-protocol effect

**Antibiotic Susceptibility Testing*

STATISTICAL APPROACH

- Features included were patient characteristics, pathogen-specific risk factors, and clinical severity metrics. **(Table 1, Figure 2)**
- Propensity scores estimated using machine learning methods including logistic regression, LASSO regression, gradient-boosted decision trees, random forest, and ensemble learning **(Table 2)**
- Marginal effects estimated via marginal structural Cox proportional hazards models in three methods: unadjusted, IPW, doubly robust **(Figure 3, Table 3)**

RESULTS: POPULATION SUMMARY

Figure 1: Flowchart of study inclusion

n=49581 Hospital encounters with blood cultures → **n=2019 (4.1%)** Hospital-Onset Sepsis Included

Adequate Antibiotics (Hours after Enrollment)	**Inadequate Antibiotics** (Hours after Enrollment)
6 Hours: 1249 (62%)	6 Hours: 1249 (62%)
12 Hours: 1460 (72%)	12 Hours: 1460 (72%)
18 Hours: 1561 (77%)	18 Hours: 458 (23%)
24 Hours: 1642 (81%)	24 Hours: 377 (19%)
36 Hours: 1696 (84%)	36 Hours: 323 (16%)
48 Hours: 1725 (85%)	48 Hours: 294 (15%)

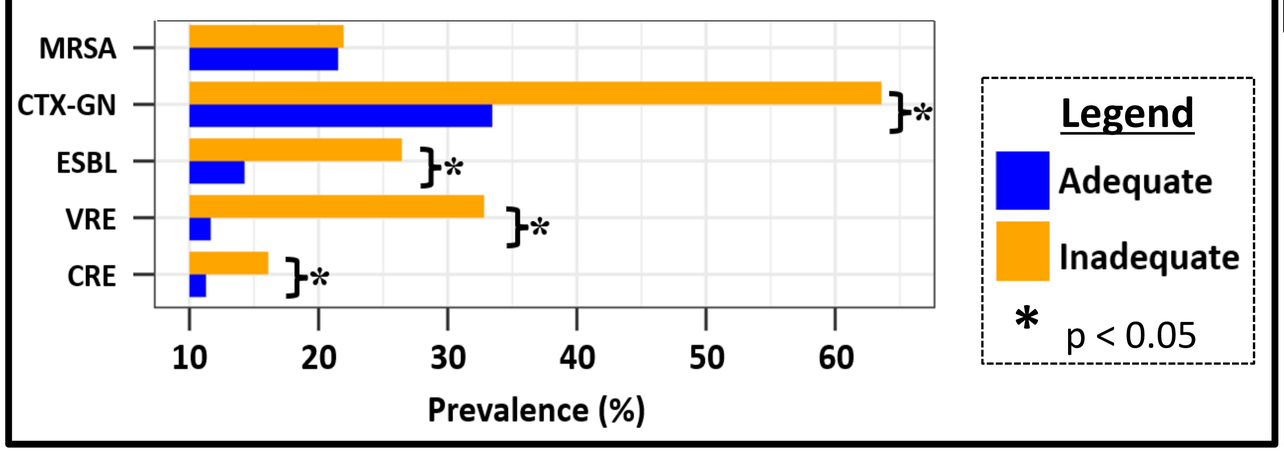
Table 2: Covariate Balance by method (SMD<0.05)

Method	Covariates Balanced
Unadjusted	53/77 (69%)
Full Logistic	75/77 (97%)
Full Logistic-Trim	**77/77 (100%)**
Gradient Boost	56/77 (72%)
Random Forest	55/77 (71%)
Ensemble Learning	61/77 (79%)

Table 1: Descriptive Summary by Treatment Group

Color of P-value indicates direction of effect	Antibiotics at 24 Hours: Adequate (n=1642)	Antibiotics at 24 Hours: Inadequate (n=377)	P Value
Demographics:			
Age, years ± SD	65.4 ± 16.5	65.1 ± 15.6	0.747
Sex, Female	714 (43.5)	186 (49.3)	0.045
Race, Black	401 (24.4)	107 (28.4)	0.134
Past Med. History:			
Chronic Pulmonary	652 (39.7)	172 (45.6)	0.040
Cerebrovascular	387 (23.6)	108 (28.6)	0.045
Hemiplegia/Paraplegia	107 (6.5)	40 (10.6)	0.008
Clinical Factors			
Vasopressor Initiation	982 (59.8)	193 (51.2)	0.003
Source of Infection			
Intra-Abdominal	313 (19.1)	97 (25.7)	0.005
Bone/Joint	113 (6.9)	16 (4.2)	0.076
Genitourinary	400 (24.4)	133 (35.3)	<0.001
Antimicrobials			
Vancomycin	1283 (78.1)	204 (54.1)	<0.001
Cefepime	1063 (64.7)	196 (52)	<0.001
Metronidazole	530 (32.3)	116 (30.8)	0.614
Piperacillin-Tazobactam	371 (22.6)	64 (17)	0.020
Ceftriaxone	162 (9.9)	58 (15.4)	0.003

Figure 2: Prevalence of resistance class by empiric antibiotics at 24 hours



RESULTS: CAUSAL MODELING

Figure 3: Mortality probability curves by treatment groups at 24 hours

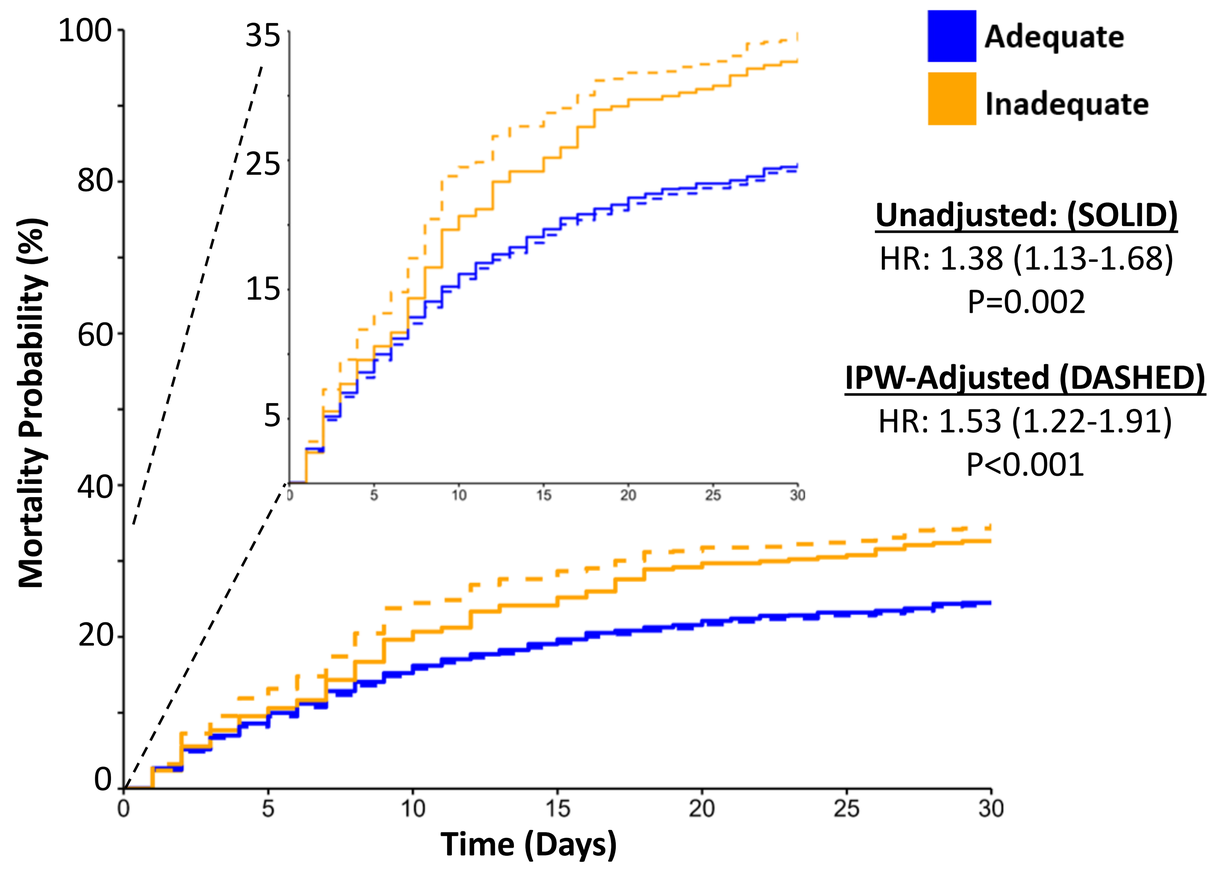


Table 3: Effect of inadequate empiric antibiotic therapy for each time

Time from Enrollment	Unadjusted HR (95%CI)	IPW-Adjusted HR (95%CI)	Doubly Robust HR (95%CI)
6 Hours	1.12 (0.94-1.33)	1.38 (1.12-1.69)	1.53 (1.24-1.90)
12 Hours	1.13 (0.94-1.36)	1.40 (1.13-1.73)	1.64 (1.31-2.06)
18 Hours	1.25 (1.03-1.52)	1.47 (1.18-1.83)	1.76 (1.40-2.22)
24 Hours	1.38 (1.13-1.68)	1.53 (1.22-1.91)	1.84 (1.45-2.32)
P for Trend	P=0.044	P=0.019	P=0.005

CONCLUSION

1. **We highlight the value of integrating real-world data sources, such as electronic health records and antibiotic susceptibility testing, with expert-driven modeling for causal effect estimation in complex clinical scenarios**
2. **One in five patients with hospital-onset sepsis received inadequate empiric antibiotic therapy due to AMR. Delays in adequate antimicrobial treatment up to 24 hours after blood culture collection caused increasing mortality.**